GPQA Diamond Benchmark Report

AgentOpt Model Selection Framework

9 Bedrock Models | 198 Samples | LangGraph Agent | March 2026

1. Brute Force Ground Truth (LangGraph)

- Exhaustive evaluation: all 9 models x 198 GPQA Diamond samples.
- Single-agent architecture: one LLM answers graduate-level science questions.
- Best model: Claude Opus 4.6 at 80.30% accuracy.
- Total brute force cost across all models: $4.13.
- Total evaluations: 1,759 of 1,782 (some samples skipped due to API errors).

Rank	Model	Accuracy	Avg Lat/Sample	Total Cost	Cost/Sample
1	**Claude Opus 4.6**	**80.30%**	86.4s	$2.43	$0.0123
2	Kimi K2.5	72.02%	97.2s	$0.72	$0.0036
3	gpt-oss-120b	68.02%	88.5s	$0.19	$0.0009
4	Claude Haiku 4.5	60.51%	83.1s	$0.51	$0.0026
5	gpt-oss-20b	52.02%	85.7s	$0.13	$0.0007
6	Qwen3 Next 80B A3B	51.04%	90.0s	$0.06	$0.0003
7	Qwen3 32B	46.67%	88.3s	$0.04	$0.0002
8	Claude 3 Haiku	37.31%	80.1s	$0.06	$0.0003
9	Ministral 3 8B	36.87%	84.7s	$0.01	$0.0000

2. Offline Selector Simulation (50 Seeds)

- Per-sample scores from brute force replayed with each selector's logic
  -- zero additional API calls required.
- Each selector simulated 50 times with different random seeds.
- Hill climbing uses topology-aware neighbors (quality/speed ranking).
  HC (1) = 1 random restart; HC (3) = 3 restarts (library default).
- Metrics: Find Rate = % of runs finding the true best model.
  Mean Acc = average true accuracy of the selector's chosen model.
  Cost = average API cost per selector run. Savings = vs brute force.

Selector	Found Best	Mean Acc	Evaluations	Cost	Savings
Brute Force	**100%**	**80.30%**	**1,759**	**$4.13**	**--**
LM Proposal	100%	80.30%	198	$2.43	41%
Arm Elimination	**98%**	**80.14%**	**444**	**$2.10**	**49%**
Hill Climbing (3)	**92%**	**79.64%**	**1,552**	**$3.75**	**9%**
Epsilon LUCB	90%	79.47%	361	$2.32	44%
Hill Climbing (1)	80%	78.65%	884	$2.78	33%
Bayesian Opt.	56%	75.41%	976	$2.32	44%
Random Search	36%	70.53%	587	$1.53	63%
Threshold SE	36%	65.88%	294	$0.26	94%

3. Key Takeaways

Brute Force Insights

- Claude Opus 4.6 leads at 80.30%, 8+ points above Kimi K2.5 (72.02%).
- gpt-oss-120b is the best value: 68.02% at only $0.19 (12.8x cheaper than Opus).
- Clear quality tiers: Opus (80%) > Kimi/gpt-oss-120b (68-72%) > Haiku 4.5 (60%) > gpt-oss-20b/Qwen3 Next (51-52%) > Qwen3 32B/C3 Haiku/Ministral (37-47%).
- Latency is similar across models (80-97s) -- Bedrock infrastructure overhead (routing, inference profiles, network) dominates actual generation time. This is specific to GPQA's simple single-call task (short prompts, short outputs). BFCL shows a similar pattern (204-228s), but multi-turn benchmarks with varying call counts per sample (e.g., HotpotQA) would show more latency variation.

Selector Insights

- Arm Elimination is the clear winner: 98% find rate at 49% cost savings. Uses only 444 of 1,759 evaluations to reliably identify the best model.
- Hill Climbing (3) is strong at 92% find rate, but only 9% savings -- 1-tuple search space (9 models) is too small for topology to save much.
- Epsilon LUCB is a strong runner-up: 90% find rate, fewest evals (361).
- Hill Climbing (1) still achieves 80% find rate with 33% savings.
- Threshold SE is ultra-cheap ($0.26, 94% savings) but unreliable (36% find rate).
- Bayesian Optimization underperforms here (56%) -- small discrete search space is a poor fit for GP-based surrogate modeling.
- LM Proposal is perfect on GPQA (100% find rate): GPT-4.1 reliably picks Opus when shown graduate-level science questions. Only benchmark where it succeeds.

Generated by AgentOpt | March 2026 | AWS Bedrock Inference Profiles